[0006] Through the execution of the vehicle-exit assist control for the target determined as the interference target as described above, it is possible to reduce a possibility that the door or the occupant comes in contact with the interference target. Meanwhile, when this control is executed even in a scene in which the vehicle-exit assist control is not required, there is a possibility that the occupant is annoyed, and there is also a possibility that the safe vehicle-exit of the occupant is conversely interfered. Here, accuracy of the target information may decrease depending on detection precision of the target information acquisition device, resulting in an error in the calculation result of the moving direction of the target. As a result, even when the target does not actually intersect with the intersecting determination line after continuing to move along the current moving direction, there may be a case where the target is determined to be an interference target that passes through the intersecting determination line within the predetermined period. That is, "the scene in which the vehicle-exit assist control is not required" is such a scene that a target that does not actually corresponds to an interference target is erroneously detected as an interference target due to the occurrence of the error in the calculation result of the moving direction of the target. Thus, it is required to suppress the unnecessary operation of the vehicle-exit control, to thereby increase reliability of this control.
The US patent application no. 2019/0299986 discloses a vehicle-exit assist apparatus having a control ECU executing preliminary controls for preventing an occupant from opening doors of the vehicle when an object is predicted to reach specific virtual line segments.

Summary of Invention

[0007] The present invention has been made in view of the above-mentioned problem. That is, one object of the present invention is to provide a vehicle-exit assist apparatus capable of suppressing unnecessary operation of vehicle-exit assist control.

[0008] According to at least one embodiment of the present invention (hereinafter "the present invention apparatus"), there is provided a vehicle-exit assist apparatus as defined in claim 1 including: a target information acquisition device (12) configured to detect a target existing on a rear side of an own vehicle, and to acquire, as target information, information on the detected target; and a control unit (10) configured to execute vehicle-exit assist control of assisting an occupant of the own vehicle in safely exiting the own vehicle. The control unit (10) is configured to: determine, based on the target information, whether an interference target which is liable to interfere with the safe exit of the occupant from the own vehicle during a vehicle stop is detected; determine whether a vehicle-exit assist condition is satisfied, the vehicle-exit assist condition being satisfied at least when the interference target is detected; determine,

when the vehicle-exit assist condition is satisfied, whether a specific condition is satisfied, the specific condition being satisfied when at least a part of the interference target exists in a rear-side vehicle width region (Rvw) being a region between a virtual line (L1) extending from a left rear corner portion of the own vehicle toward a rear direction

(-x-axis direction) in a front-and-rear direction (x-axis direction) of the own vehicle and a virtual line (L2) extending from a right rear corner portion of the own vehicle toward the rear direction (-x-axis direction); execute the vehicle-exit assist control when the specific condition is not satisfied; and avoid executing the vehicle-exit assist control when the specific condition is satisfied.

[0009] The related-art vehicle-exit assist apparatus executes the vehicle-exit assist

control, for example, when an execution condition, which is satisfied when the interference target is detected during the vehicle stop and the vehicle-exit intention of an occupant is detected, is satisfied. With this configuration, the vehicle-exit control is executed even when a target that does not actually correspond to an interference target is erroneously detected as an interference target due to an error in a calculation result of a moving direction of the target and the execution condition is consequently satisfied. Such execution of the vehicle-exit control thus causes unnecessary operation.

[0010] Here, the inventors of the present invention have found that when a target is erroneously detected as an interference target due to the occurrence of the error in the calculation result of the moving direction of the target, the target tends to be moving in a region extending on a directly rear side of the own vehicle. Based on this fining, when the vehicle-exit assist condition is satisfied, the present invention apparatus is configured not to immediately execute the vehicle-exit assist control, but to determine whether or not the specific condition, that is, the condition which is satisfied when at least a part of the interference target is positioned in the rear-side vehicle width region of the own vehicle, is satisfied. When the specific condition is not satisfied, the present invention apparatus is configured to determine that it is less liable that the target is erroneously detected as the interference target, and to execute the vehicle-exit assist control. When the specific condition is satisfied, the present invention apparatus is configured to determine that it is highly liable that the target is erroneously detected as the interference target due to the occurrence of the error in the calculation result of the moving direction of the target, and to avoid executing the vehicle-exit assist control.

[0011] With this configuration, the unnecessary operation of the vehicle-exit assist control can be suppressed while this control is appropriately executed in a scene in which this control is actually required, thereby being capable of increasing reliability of the vehicle-exit assist control.

[0012] According to at least one aspect of the present invention, and as defined in claim 1, of a front end portion of the interference target, a portion closest to the own vehicle in the vehicle width direction of the own vehicle is defined as a proximate portion (np), virtual lines that each have a predetermined length, are positioned on a reference axis (y-axis) passing through the left rear corner portion and the right rear corner portion of the own vehicle, and extend toward vehicle width outward directions from the left rear corner portion and the right rear corner portion, respectively, are defined as intersecting determination lines (LL, LR), and the control unit (10) is configured to: determine that the detected target is an interference target when an

intersection between an extension line extending from the proximate portion (np) along a moving direction of the detected target and the reference axis (y-axis) is positioned on one of the intersecting determination lines (LL, LR), and a predicted period (TTC) predicted to be required for the detected target to reach the intersection is equal to or shorter than a predetermined period threshold value (TTCth); and determine that the specific condition is satisfied further when a magnitude (|Δy|) of a change amount per unit period of a position of the intersection is equal to or larger than a predetermined change amount threshold value (Δyth).

[0013] The inventors of the present invention have found that when a target is erroneously detected as an interference target due to the occurrence of the error in the calculation result of the moving direction, the magnitude of the change amount per unit period of the position of the intersection between the extension line extending from the proximate portion along the moving direction of the target and the reference axis tends to be relatively large. Based on this finding, the at least one aspect of the present invention is configured such that the specific condition is satisfied further when the magnitude of the change amount per unit period of the position of the intersection is equal to or larger than the change amount threshold value. With this configuration, the precision of the determination of whether or not the vehicle-exit control is the unnecessary operation becomes higher, and the vehicle-exit assist control can thus more appropriately be executed.

[0014] According to at least one aspect of the present invention, and as defined in claim 1; the control unit (10) is configured to determine that the specific condition is satisfied further when a component (vx) of a speed (v) of the interference target in the front-and-rear direction of the own vehicle is higher than 0 and is lower than a predetermined speed threshold value (vxth) and/or a component (dx) of a deceleration (d) of the interference target in the front-and-rear direction is equal to or higher than a predetermined deceleration threshold value (dxth).

[0015] The inventors of the present invention have found that when a target is erroneously detected as an interference target due to the occurrence of the error in the calculation result of the moving direction, the target tends to move at such a speed that "the component thereof in the front-and-rear direction of the own vehicle" has a relatively small value, and/or at such a deceleration that the component thereof in the front-and-rear direction has a relatively large value. Based on this finding, the at least one aspect of the present invention is configured such that the specific condition is satisfied further when the component of the speed of the interference target in the front-and-rear direction is higher than 0 and lower than the speed threshold value and/or